KENYATTA UNIVERSITY

CIVIL ENGINEERING DEPARTMENT

ECV 207: FLUID MECHANICS II

CONTINUOUS ASSESSMENT TEST- answer both questions

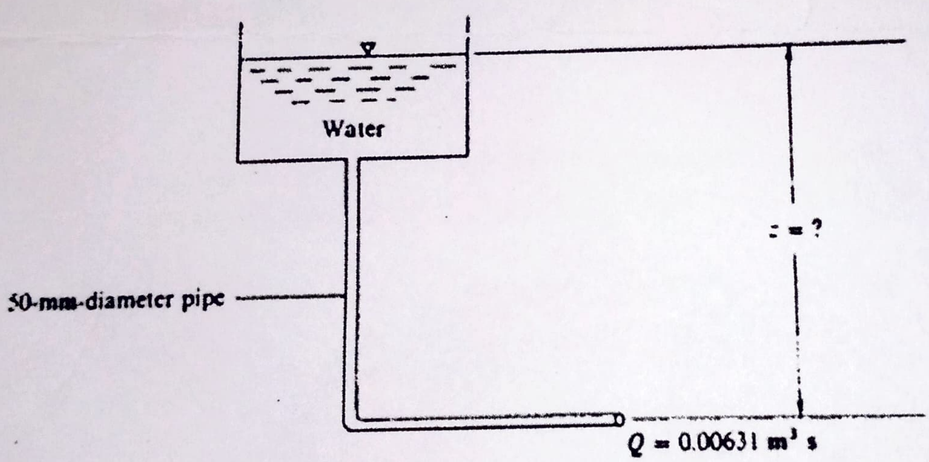
Time: 1hr

Question 1

i) A pipe measuring 100m long has a Manning's roughness coefficient 'n' of 0.012 and C_{HW} =120. Compare the headloss calculated using both the Manning's equation and Hazen-William's equation (5mks).

$v = 0.849\, C_{HW}\, R^{0.63} S^{0.54}$

ii) Water is to be delivered from a reservoir through a pipe to a lower level and discharged into the air as shown in the figure below. If the head loss in the entire system is **11.58m**, determine the vertical distance between the point of water discharge and the water surface in the reservoir (5mks)



Question 2

A fluid flows through a **10mm** diameter tube at a Reynolds' Number of **1800**. The loss is **30m** in a **120m** length of tubing. Calculate the discharge in litres per minute (5m

KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECV 207 : FLUID MECHANICS II

1ST SEMESTER 2021/2022 TIME: 2 Hours

INSTRUCTIONS:

(i) QUESTION ONE IS COMPULSORY

(ii) ATTEMPT ANY OTHER TWO QUESTIONS

(iii) REASONABLE ASSUMPTIONS SHOULD BE MADE WHERE NECESSARY

(iv) RELEVANT TABLES AND CHARTS HAVE BEEN ATTACHED

Question 1

i) Describe the water hammer phenomenon in pipelines. Suggest three ways in which the detrimental impacts of water hammer can be eliminated in pipelines (5mks)

ii) Figure Q1(ii) shows a looping pipe system. Pressure heads at points A and E are 70.0m and 46.0m, respectively. Compute the flow rate of water through each branch of the loop. Assume C=120 for all pipes

(15mks)

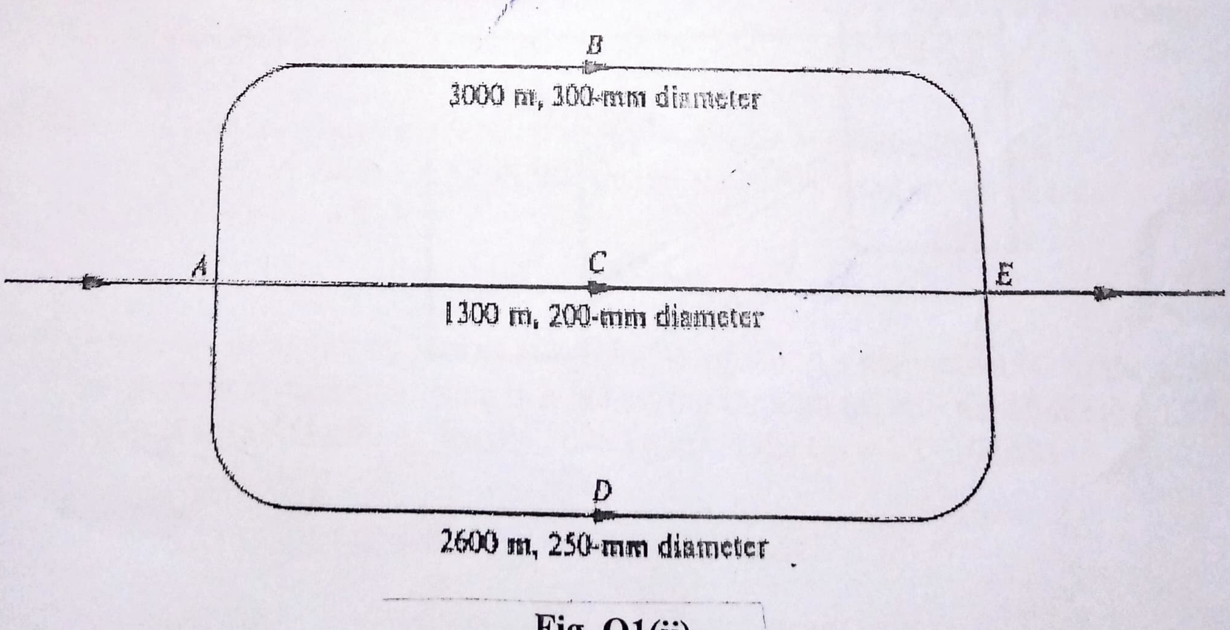


Fig. Q1(ii)

iii) Water is flowing through capillary tubes A and B into tube C as shown in Fig Q1(iii). If QA =3mL/s in tube A, what is the largest Q_B allowable in tube B for laminar flow in tube C? The water is at a temperature of 40^0C. With the calculated Q_B, what kind of flow exists in tubes A and B?

(10mks)

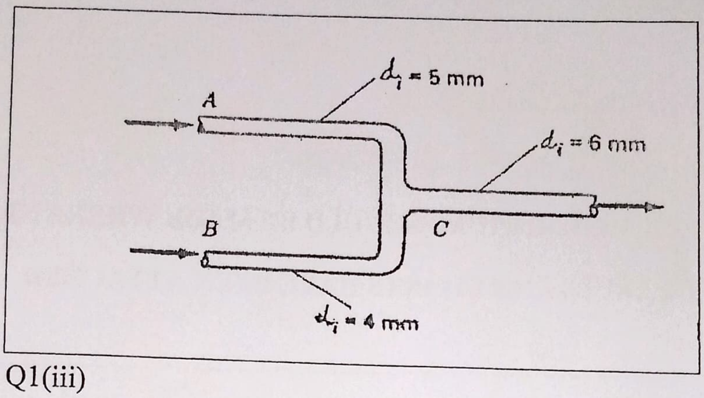


Fig Q1(iii)

__Question 2__

The reservoirs in Fig Q2 contain water at 20°C. If the pipe is smooth with L=7km and d=50mm, what will the flow rate be for Δz=98m by using an iterative procedure for determining the value of the coefficient of friction? Relevant tables and charts are provided.

(20mks)

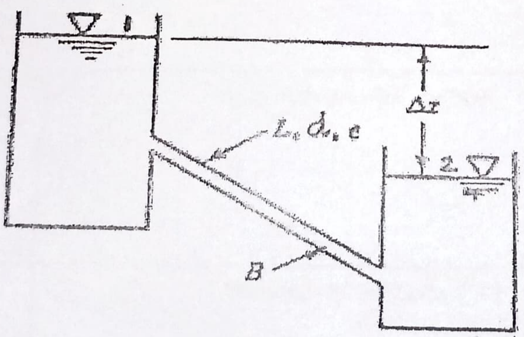


Fig Q2

Question 3

$Re = \frac{DV}{\nu} = \frac{DV}{\mu/\rho} = \frac{\rho D V}{\mu}$

i) For laminar flow in a circular pipe, verify that the pipe frictional coefficient can be expressed by the following equation:

$\lambda = 64/Re$ (10mks)

$\frac{V_1^2}{2g} + \frac{P_1}{\rho g} + Z_1 = \frac{V_2^2}{2g} + \frac{P_2}{\rho g} + Z_2 + h_L$

$\frac{P_1 - P_2}{\rho g} = \frac{P_1 - P_2}{\gamma} = h_L$ $\quad \frac{(P_1 - P_2) D^2}{32 L}$ $\quad = \frac{64 \mu}{\rho D V}$ $\quad \lambda = 64/Re$ HW

ii) A fluid flows through a 10mm diameter tube at a Reynolds' Number of 1800. The head loss is 30m in a 120m length of tubing. Calculate the discharge in litres per minute (5mks)

iii) A smooth flat plate 2.0m wide and 10.0m long is being towed lengthwise through still water at 20°C at a velocity of 5.0m/s. What is the drag force on one side of the plate, i.e. the skin-friction drag? (5mks)

$F_D = C_D \left(\frac{V^2}{2}\right) \rho A$

Question 4

i) Water at 20^0C flows in a 100mm diameter new cast iron pipe with a velocity of 5.0 m/s. Determine the power lost to friction. Take e = 0.00026 for new cast iron pipes

$h_f = \frac{4f L V^2}{2Dg}$ (10mks)

iv) A 2.0m diameter, 1600m long concrete pipe (e=1.5mm) carries water at 12^0C between two reservoirs at $8m^3/s$. Find the difference in water surface elevation between the two reservoirs, considering minor losses at entrance and exit (10mks)

Question 5

i) Discuss the development of a boundary layer for a flow around a solid body according to Ludwig Prandtl (5mks)

ii) Whenever a body is placed in a flow, the body is subject to a force from the surrounding fluid. With a suitable illustration show how drag and lift forces develop in a body placed in a flow (5mks)

iii) A flat plate, 0.1 m by 0.1 m, moves at a velocity of 5.0 m/s perpendicular to the plate. Find the drag force on the plate if it is moving through (a) still air of density 1.200 kg/m^3 and (b) still water of density 1000 kg/m^3. Take $C_D = 1.1$ (10mks)

In the early 1940s, it was somewhat cumbersome to use any of these implicit equations in engineering practice. A convenient chart was prepared by Lewis F. Moody* (Figure 3.8); it is commonly called the *Moody diagram of friction factors for pipe flow.*

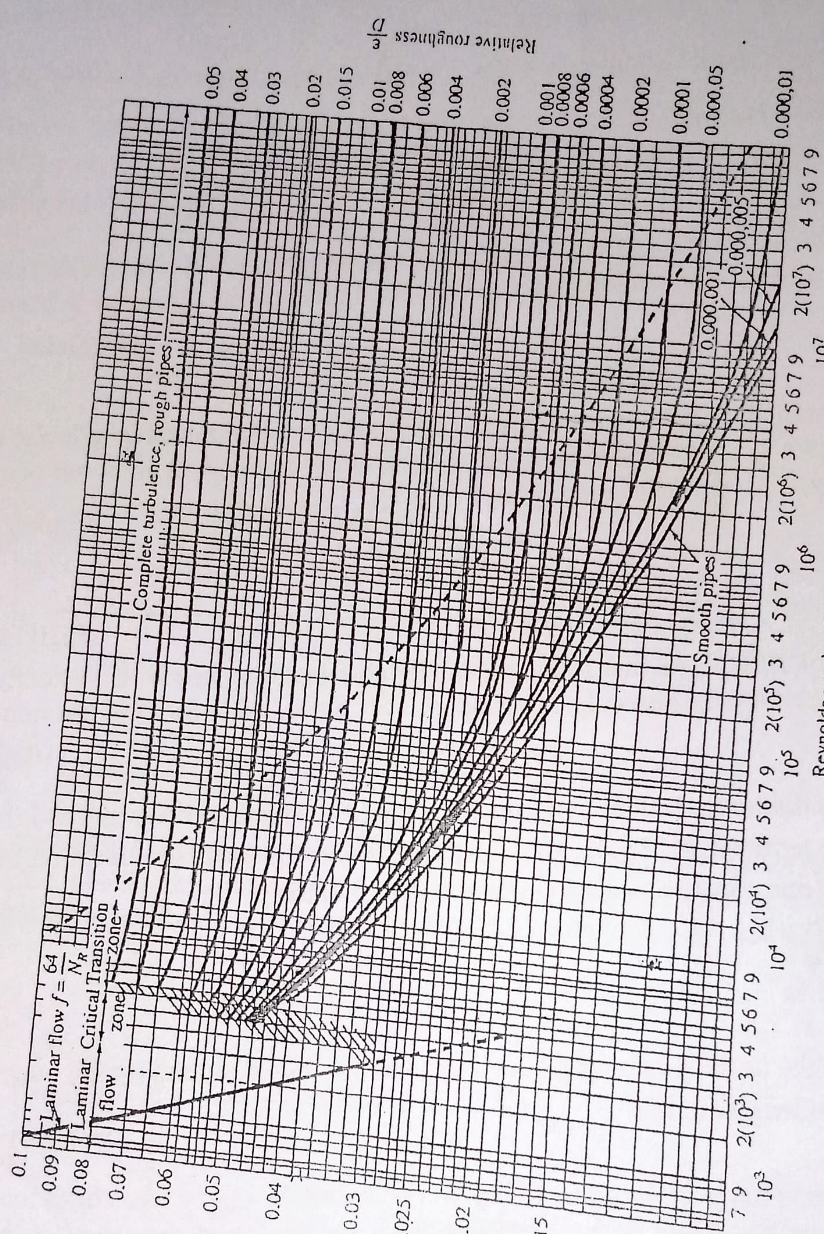


Figure 3.8 Friction factors for flow in pipes: the Moody diagram. *Source:* From L. F. Moody, "Friction factors for pipe flow," *Trans. ASME*, vol. 66, 1944.

* L. F. Moody, "Friction factors for pipe flow," *Trans. ASME*, 66 (1944).

INVOLVEMENT IN ***EXAMINATION IRREGULARITY*** *SHALL LEAD TO* ***DISCONTINUATION***

Table of Water and Air Viscosity with Temperature

TABLE 1.3 Viscosities of Water and Air

Temperature (°C)	Water: Viscosity (μ) $N \cdot sec/m^2$	Water: Kinematic Viscosity (ν) m^2/sec	Air: Viscosity (μ) $N \cdot sec/m^2$	Air: Kinematic Viscosity (ν) m^2/sec
0	1.781×10^{-3}	1.785×10^{-6}	1.717×10^{-5}	1.329×10^{-5}
5	1.518×10^{-3}	1.519×10^{-6}	1.741×10^{-5}	1.371×10^{-5}
10	1.307×10^{-3}	1.306×10^{-6}	1.767×10^{-5}	1.417×10^{-5}
15	1.139×10^{-3}	1.139×10^{-6}	1.793×10^{-5}	1.463×10^{-5}
20	1.002×10^{-3}	1.003×10^{-6}	1.817×10^{-5}	1.509×10^{-5}
25	0.890×10^{-3}	0.893×10^{-6}	1.840×10^{-5}	1.555×10^{-5}
30	0.798×10^{-3}	0.800×10^{-6}	1.864×10^{-5}	1.601×10^{-5}
40	0.653×10^{-3}	0.658×10^{-6}	1.910×10^{-5}	1.695×10^{-5}
50	0.547×10^{-3}	0.553×10^{-6}	1.954×10^{-5}	1.794×10^{-5}
60	0.466×10^{-3}	0.474×10^{-6}	2.001×10^{-5}	1.886×10^{-5}
70	0.404×10^{-3}	0.413×10^{-6}	2.044×10^{-5}	1.986×10^{-5}
80	0.354×10^{-3}	0.364×10^{-6}	2.088×10^{-5}	2.087×10^{-5}
90	0.315×10^{-3}	0.326×10^{-6}	2.131×10^{-5}	2.193×10^{-5}
100	0.282×10^{-3}	0.294×10^{-6}	2.174×10^{-5}	2.302×10^{-5}

Minor Losses Coefficients

Obstruction	K
tank exit	0.5
tank entry	1.0
smooth bend	0.30
Mitre bend	1.1
Mitre bend with guide vanes	0.2
90 degree elbow	0.9
45 degree elbow	0.42
Standard T	1.8
Return bend	2.2
Strainer	2.0
Globe valve, wide open	10.0
Angle valve, wide open	5.0
Gate valve, wide open	0.19
$^3/_4$ open	1.15
$^1/_2$ open	5.6
$^1/_4$ open	24.0
Sudden enlargement	0.10
Conical enlargement: 6°	0.13
(total included angle) 10°	0.16
15°	0.30
25°	0.55
Sudden contractions:	
area ratio 0.2	0.41
(A_2/A_1) 0.4	0.30
0.6	0.18
0.8	0.06

Table 2.1: Obstruction Losses in Flow Systems

LIST OF SURFACES WITH ROUGHNESS $k = 0{\cdot}6$mm

GOOD examples of:

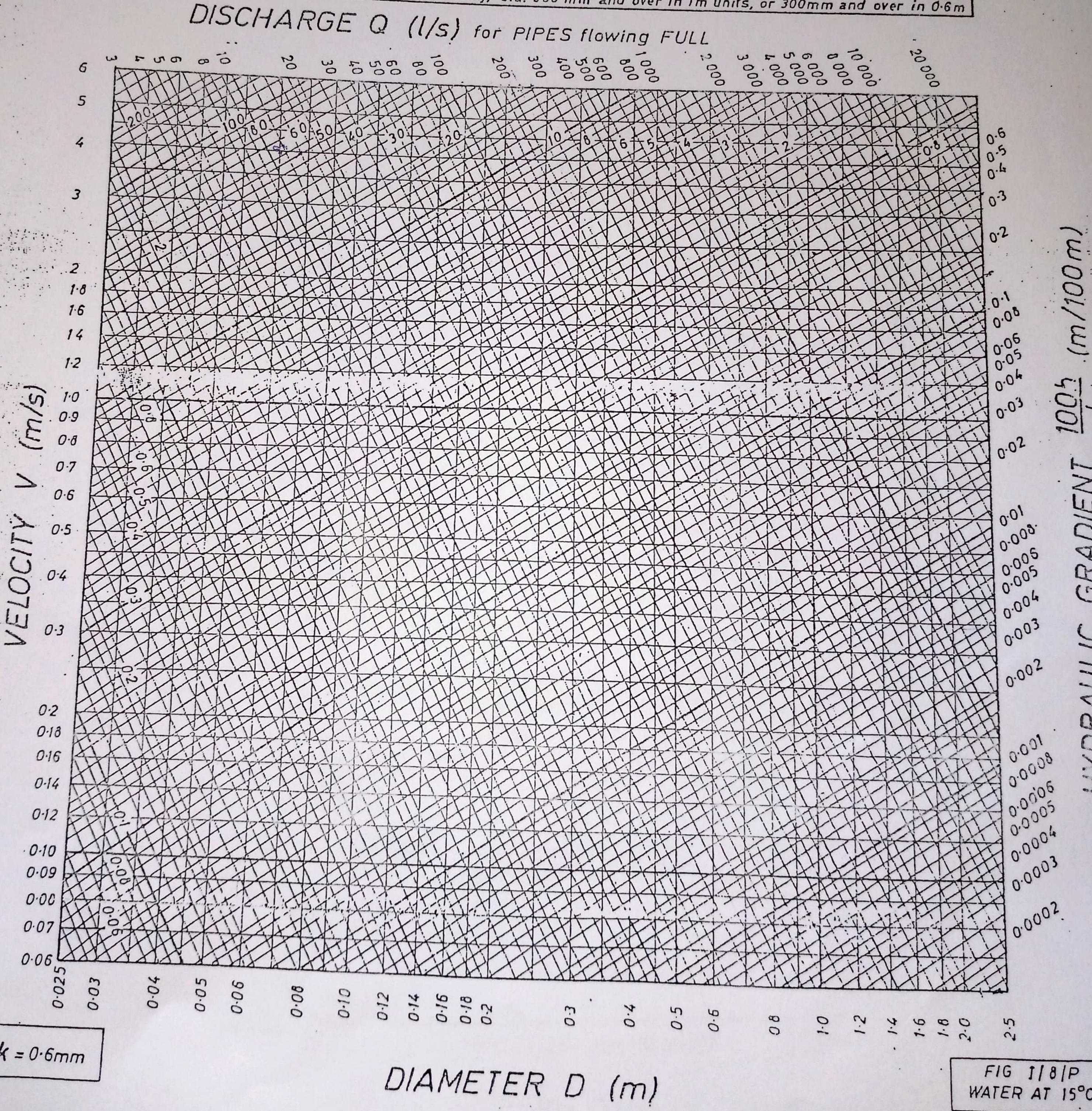
Riveted steel pipes, untuberculated, 6mm or under plates (girth riveted only).
Water mains slightly attacked by tuberculation (up to 20 years use).
Concrete class 2. Monolithic construction against rough forms, rough texture precast pipes, or cement gun surface (for very coarse textures, take k = size of aggregate in evidence).
Butt jointed drain tile.
Glazed brickwork.

NORMAL examples of:

Mature foul sewers, slimed to not more than 6mm.
Rusty wrought iron.
Wood stave pipes planed plank flumes.
Concrete class 3. Monolithic construction against steel forms, wet-mix or spun precast pipes, or with cement or asphalt coating.
Smooth trowelled surfaces.

POOR examples of:

Uncoated cast iron pipes.
Tate relined pipes.
Glazed vitrified clay, dia. 600 mm and over in 1m units, or 300mm and over in 0·6m



k = 0·6mm

FIG I/8/P
WATER AT 15°C